

Learning a Safety-Aware CNN Cost Map for A*-Based Path Planning in Simulated Autonomous Driving Scenarios

Midpoint Check Paper Draft

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Draft status. The Abstract, Introduction, Related Work, Research Project Problem, and Method sections are near-complete. The current prototype includes runnable scene generation, standard A*, a safety-aware expert planner, quantitative metrics, Gaussian label generation, and train/validation/test dataset generation. Final CNN training and full hypothesis testing are explicitly marked as placeholders.

Abstract

Classical A* search provides an interpretable and reliable solution to grid-based path planning, but its behavior depends strongly on the cost representation supplied to the search. A purely geometric cost tends to favor the shortest collision-free route and does not directly encode driving preferences such as obstacle clearance, lane-centered motion, or road-boundary compliance. This project investigates a hybrid learning-and-search framework in which a lightweight convolutional encoder-decoder predicts a path-preference or residual cost map from a simplified bird's-eye-view driving scene, while a standard A* planner remains responsible for producing the final collision-free route. The current midpoint prototype generates 64 x 64 synthetic road scenes with five input channels, produces expert demonstrations using a rule-based safety-aware A* planner, converts those paths into Gaussian soft labels, and constructs non-overlapping training, validation, and test splits of 800, 100, and 100 samples. Preliminary evaluation over 100 scenes shows that the safety-aware expert preserves a 100% planning success rate, increases mean minimum obstacle clearance from 1.340 to 2.438 grid cells, and reduces mean lane deviation from 1.628 to 1.197 grid cells, while increasing average path length by only 3.1%. The final stage will train a lightweight U-Net and evaluate whether the learned guidance map can reproduce these safety improvements while maintaining the transparency and feasibility guarantees of search-based planning.

Keywords: A* search; path planning; learned cost map; U-Net; autonomous driving; synthetic grid benchmark; safety-aware planning

1. Introduction

Path planning is a central capability in autonomous robots and automated vehicles. A planner must connect a start state to a goal while respecting obstacles and other environmental constraints. In a discrete grid representation, A* search is attractive because it is conceptually transparent, easy to verify, and optimal when its heuristic and cost assumptions are satisfied (Hart et al., 1968). However, the behavior of A* is determined by the objective it optimizes. When the traversal cost includes only geometric distance and hard obstacle occupancy, the resulting route may pass very close to obstacles, deviate unnecessarily from a lane center, or exhibit behavior that is valid in a graph-theoretic sense but undesirable for driving.

Autonomous-driving path quality is therefore not adequately described by collision avoidance alone. A useful route should maintain clearance from static obstacles, remain inside the drivable road region, prefer lane-centered motion when possible, and avoid excessive detours or abrupt local direction changes. These preferences can be encoded manually as weighted cost terms, but hand-tuned cost functions often become brittle as scene complexity grows. Conversely, a neural network can learn spatial preferences from demonstrations, but a network that directly outputs an action sequence or path may produce disconnected or unsafe predictions and can be difficult to interpret.

This project adopts a hybrid strategy. A convolutional neural network predicts a dense path-preference map from a simplified bird's-eye-view scene, and a conventional A* planner uses that prediction as a soft spatial cost while continuing to enforce hard obstacle and road-boundary constraints. The network therefore does not

replace search. Instead, it supplies a learned representation that guides an interpretable planner. This design is related to prior work on learned cost functions, differentiable planning, and neural guidance for graph search (Wulfmeier et al., 2017; Yonetani et al., 2021).

The work is intentionally evaluated first in a controlled synthetic benchmark. Procedurally generated grid worlds make it possible to vary road curvature, obstacle placement, and scene difficulty while preserving reproducibility through random seeds. Grid-based benchmarks are well established for pathfinding research (Sturtevant, 2012), and synthetic environments have also been used to test whether learned planners generalize to unseen layouts (Tamar et al., 2016; Yonetani et al., 2021). The purpose of the benchmark is not to claim immediate real-world driving readiness. Rather, it is to isolate the central research question: can a learned dense cost representation improve the safety-related behavior of A* without sacrificing basic feasibility and efficiency?

2. Related Work

2.1 Classical Grid Search and Benchmarks

Hart, Nilsson, and Raphael formalized A* as a heuristic graph-search method that combines the exact cost accumulated from the start with an estimate of the remaining cost to the goal (Hart et al., 1968). For an 8-connected grid with unit axial moves and square-root-of-two diagonal moves, octile distance provides a natural geometric heuristic. Standardized grid benchmarks have subsequently supported reproducible comparison of path length, runtime, and search effort across algorithms (Sturtevant, 2012). These works motivate the use of A* as the interpretable search backbone and the use of fixed synthetic test sets for controlled evaluation.

2.2 Learning Cost Maps and Planning Preferences

Learning a spatial cost function from demonstrations offers an alternative to manually selecting every planning penalty. Wulfmeier et al. (2017) learned large-scale navigation costs from expert driving demonstrations using deep inverse reinforcement learning. Their work supports the idea that driving preferences can be represented as a dense cost map rather than as direct low-level control. Pérez-Higueras et al. (2018) similarly used a fully convolutional network to predict a feasible path representation from expert demonstrations and then combined the prediction with a conventional sampling-based planner. The broader principle is directly relevant to this project: a learned map can express behavior preferences, while a classical planner can provide connectivity and collision checking.

2.3 Neural Planning and Learned Search Guidance

Value Iteration Networks embed an approximate planning computation inside a differentiable neural architecture and demonstrate generalization on path-planning domains (Tamar et al., 2016). Learning Heuristic Search via Imitation trains search guidance from expert or oracle decisions to reduce unnecessary search effort (Bhardwaj et al., 2017). Neural A* couples a convolutional encoder with a differentiable reformulation of A* and learns a guidance map from expert paths (Yonetani et al., 2021). The present project is simpler than fully differentiable A*: the planner itself remains a conventional implementation, and only the spatial preference map is learned. This separation reduces implementation risk for a course-scale project and preserves a clear distinction between learned soft guidance and hard planning constraints.

2.4 U-Net as a Dense Prediction Model

The network output in this project is a dense 64 x 64 map rather than a single class label. U-Net is a suitable baseline because its contracting path captures global context while its expanding path and skip connections preserve spatial localization (Ronneberger et al., 2015). A lightweight U-Net can therefore map the five-channel scene tensor to a one-channel path-preference map without requiring a large model or an extensive real-world dataset.

3. Research Project Problem

3.1 Problem Statement

Given a simplified bird's-eye-view grid containing a drivable-road mask, static obstacles, lane-center information, a start location, and a goal location, the objective is to generate a connected and collision-free path that balances geometric efficiency with driving-oriented safety preferences. Standard geometric A* serves as the primary baseline. The proposed learned method predicts a dense map that lowers the effective traversal cost in regions resembling expert behavior and raises the cost elsewhere. Obstacles and cells outside the road remain non-traversable regardless of the network output.

3.2 Research Questions and Hypotheses

- H1:** A CNN-predicted path-preference or cost map will guide A* to produce paths that are closer to safety-aware expert paths than paths produced by standard geometric A*.
- H2:** CNN-guided A* will improve safety-related metrics, particularly minimum obstacle clearance and lane-center adherence, relative to standard geometric A*.
- H3:** CNN-guided A* will maintain a comparable success rate and path-length ratio while improving path quality in cluttered driving-like scenes.

4. Method

Figure 1 summarizes the intended learning-and-search pipeline. The midpoint implementation has completed every stage through dataset generation; U-Net training and learned-cost integration remain for the final project phase.



Figure 1. Proposed hybrid learning-and-search pipeline.

4.1 Synthetic Driving-Scene Representation

Each sample is represented on a 64 x 64 grid. A mildly curved road centerline is generated procedurally, and the drivable road is formed by a fixed half-width around that centerline. Rectangular static obstacles are sampled within the road. The generator reserves clear regions around the start and goal and rejects obstacle layouts for which no valid path exists. An 8-connected breadth-first search is used only as a feasibility check during scene generation, and diagonal corner cutting is prohibited.

Channel	Content	Interpretation
1	Road mask	Binary drivable region
2	Obstacle mask	Binary static-obstacle occupancy
3	Lane-center preference	Gaussian preference centered on the road centerline
4	Start map	Gaussian heatmap around the start cell
5	Goal map	Gaussian heatmap around the goal cell

4.2 Standard A* Baseline

The standard baseline searches the free-space grid using axial moves of cost 1 and diagonal moves of cost square root of 2. Octile distance is used as the heuristic. The implementation returns the complete path, geometric length, number of expanded nodes, planning time, and success status. A separate path validator

checks that the route begins at the specified start, reaches the goal, remains in free space, uses only adjacent grid moves, and never cuts diagonally through obstacle corners.

4.3 Safety-Aware Expert Planner

The expert planner is a spatial-cost-aware version of A*. It uses the same hard feasibility constraints as the standard planner, but its traversal map assigns additional cost near obstacles and road boundaries and away from the lane center. The free-space cell cost is defined as:

$$C(x, y) = I + w_o C_obstacle(x, y) + w_b C_boundary(x, y) + w_l C_lane(x, y).$$

The obstacle and boundary terms decay exponentially with Euclidean distance, while the lane term is one minus the lane-center preference map. The midpoint implementation uses weights $w_o = 8$, $w_b = 4$, and $w_l = 2$. Cells outside the road and cells occupied by obstacles receive infinite cost. For a move between adjacent cells u and v , the edge cost is the movement distance multiplied by the average of $C(u)$ and $C(v)$. The heuristic is octile distance multiplied by the minimum finite cell cost, preserving a lower-bound estimate of remaining objective cost.

4.4 Expert Labels

A one-pixel-wide expert path is highly imbalanced relative to the full grid and provides little learning signal to nearby cells. The binary path is therefore converted into a Gaussian soft target. Let $d(x, y)$ denote the Euclidean distance from cell (x, y) to the nearest expert-path cell. The supervision target is:

$$P_expert(x, y) = \exp[-d(x, y)^2 / (2 \sigma^2)], \quad \sigma = 2 \text{ grid cells}.$$

The label is set to zero on obstacles and outside the drivable region. This target provides a smooth path corridor: the exact expert path has value 1, nearby cells receive intermediate values, and distant cells approach zero.

4.5 Dataset Generation and Splits

The data pipeline automatically generates a scene, constructs the rule-based safety cost map, runs expert A*, validates the resulting path, builds the Gaussian label, and stores the five-channel input, soft label, binary path mask, start and goal coordinates, random seed, and expert path length. The current midpoint dataset contains 800 training samples, 100 validation samples, and 100 test samples. The three splits use disjoint seed ranges to prevent exact scene duplication. Generated arrays are saved as compressed NumPy archives locally and are excluded from the submitted repository, while the notebook retains the code required to reproduce them.

4.6 Planned CNN and Hybrid Inference

The planned model is a lightweight U-Net that maps a tensor of shape $5 \times 64 \times 64$ to a one-channel path-preference map of shape $1 \times 64 \times 64$. A sigmoid output will constrain predictions to $[0, 1]$. The initial loss will combine pixelwise binary cross-entropy with a region-overlap or regression term if needed. At inference time, the predicted preference will be converted to a nonnegative learned cost, for example $C_learned = 1 - P_CNN$ or $-\log(P_CNN + \epsilon)$, and combined with the rule-based hard constraints. A scalar λ will control the influence of the learned term.

PLACEHOLDER - Final model specification

After implementation, insert the exact U-Net depth, base channel count, parameter count, optimizer, learning rate, batch size, epoch count, checkpoint-selection rule, and loss function. Include a compact architecture table or diagram.

5. Experimental Design

5.1 Baselines and Comparison Methods

The final evaluation will compare: (1) standard geometric A*, (2) the rule-based safety-aware expert A*, and (3) CNN-guided A*. Dijkstra search may be included as an additional uninformed-search baseline if time permits. Standard A* measures the behavior of a purely geometric planner, the expert establishes an upper reference for

the hand-defined safety objective, and CNN-guided A* tests whether the network can approximate expert preferences on unseen scenes.

5.2 Evaluation Metrics

- Success rate: fraction of scenes with a valid start-to-goal path.
- Path length: sum of Euclidean lengths of grid moves.
- Minimum and mean obstacle clearance: distance from path cells to the nearest obstacle.
- Minimum road-boundary clearance: distance from path cells to the nearest non-road cell.
- Mean and maximum lane deviation: lateral distance from each path cell to the road centerline.
- Mean and total turn angle: local heading-change measures used as path-smoothness indicators.
- Expanded nodes and planning time: measures of search effort and computational cost.
- Expert-path similarity: planned for the final study using mean nearest-path distance, Chamfer distance, or path overlap.

PLACEHOLDER - Final learned-model experiment

Insert the final CNN training curves, representative prediction maps, CNN-guided A* paths, comparison on the 100-scene test set, and at least one failure case. Report all hyperparameters and state whether the checkpoint was selected using validation loss or a planning metric.

6. Statistical Analysis Plan and Hypothesis Tests

All planners will be evaluated on the same held-out test scenes, creating paired observations. Continuous path metrics will first be summarized by mean, standard deviation, median, and a 95% bootstrap confidence interval. For each metric, the paired difference between CNN-guided A* and standard A* will be examined. A paired t-test will be used if the difference distribution is approximately symmetric and free of severe outliers; otherwise, the Wilcoxon signed-rank test will be reported. Because several metrics are tested, the final report will distinguish primary metrics (minimum obstacle clearance and mean lane deviation) from secondary metrics and will avoid interpreting isolated p-values without effect sizes and confidence intervals.

H1 will be evaluated using an expert-path similarity measure on the held-out test set. H2 will be evaluated primarily through paired changes in obstacle clearance and lane deviation. H3 will be evaluated through success rate, path-length ratio, and planning time. If success outcomes differ across methods, a paired binary comparison such as McNemar's test will be used; if all methods achieve 100% success, the result will be reported descriptively because no variation is available for a significance test.

PLACEHOLDER - Hypothesis evidence

For the final submission, insert one concise decision paragraph for H1, H2, and H3. Each decision should include the metric, effect size, confidence interval, statistical test if applicable, and whether the evidence supports, partially supports, or does not support the hypothesis.

7. Preliminary Results

Before training the CNN, the standard and expert planners were evaluated on 100 procedurally generated scenes. These results verify that the expert used to produce supervision actually represents the intended safety preference. Both methods achieved a 100% success rate. The safety-aware expert increased minimum obstacle clearance and reduced lane deviation while maintaining road-boundary clearance. The trade-off was a modest increase in path length, more local heading changes, and higher search effort.

Metric	Standard A*	Expert A*	Relative change
Success rate	1.000	1.000	No change
Path length	61.203	63.092	+3.1%
Minimum obstacle clearance	1.340	2.438	+81.9%
Minimum boundary clearance	6.803	6.883	+1.2%
Mean lane deviation	1.628	1.197	-26.5%
Mean turn angle	0.064	0.156	+143.8%
Expanded nodes	174.750	908.960	+420.1%
Planning time (ms)	0.695	4.450	+540.3%

Table 1. Preliminary comparison over 100 generated scenes. Distances are measured in grid cells; turn angle is in radians.

The preliminary results support the design of the expert-label generator. Relative to geometric A*, the expert improves the two primary safety targets: minimum obstacle clearance increases by approximately 81.9%, and mean lane deviation decreases by approximately 26.5%. The average geometric path grows by only 3.1%, and minimum boundary clearance is maintained. These effects suggest that the weighted cost map is not simply producing large detours; it is changing where the path lies within the available free space.

The expert expands substantially more nodes and takes longer to plan, although the mean runtime remains below 5 ms on the current machine. Its higher turn-angle metric indicates more local direction changes. Because the current expert does not include an explicit orientation state or turning penalty, this result is expected and will be treated as a limitation. The final learned planner will be evaluated to determine whether the CNN prediction smooths or amplifies this behavior.

8. Limitations and Threats to Validity

- The benchmark is synthetic and two-dimensional. It does not model vehicle dynamics, nonholonomic constraints, dynamic obstacles, perception uncertainty, traffic rules, or interaction with other agents.
- The expert behavior is defined by a hand-designed weighted cost function. The network will learn this operational definition of safety, not a universal or human-validated driving policy.
- The current road generator produces a limited family of mildly curved roads and rectangular obstacles. Generalization claims must therefore be restricted to the tested procedural distribution and any explicitly designed out-of-distribution scenarios.
- Grid resolution influences all distance and smoothness metrics. Values in grid cells should not be interpreted as physical meters without an explicit map scale.
- The midpoint dataset is sufficient for a feasibility study but small relative to production machine-learning datasets. Overfitting will be monitored through a disjoint validation split and held-out test seeds.
- The current turn metric is based on discrete heading changes and does not equal vehicle curvature. A future extension could apply post-processing or incorporate heading into the search state.

9. Conclusion and Remaining Work

The midpoint prototype establishes a complete and reproducible data-generation foundation for learning-guided A* planning. Synthetic driving scenes, standard A*, the safety-aware expert, path validation, quantitative metrics, Gaussian labels, and the 800/100/100 dataset splits are implemented and runnable in the draft

notebooks. Preliminary results demonstrate that the expert produces substantially safer and more lane-centered paths with only a small increase in geometric length, providing a meaningful supervision signal for the learning stage. The remaining work is to implement and train the lightweight U-Net, convert its prediction into a learned traversal cost, run CNN-guided A* on the held-out test set, perform the planned paired statistical analysis, and document representative successes and failures. The final conclusions regarding H1-H3 will be based on those learned-model experiments rather than on the rule-based expert results alone.

Reproducibility Note

The prototype implementation and draft Jupyter notebooks are available in the project GitHub repository provided on the first page of this paper. Reusable Python modules are organized in the `src/` directory, while executable notebooks are provided in the `notebooks/` directory. The current repository includes synthetic scene generation, standard A* search, safety-aware expert A* search, safety-cost-map construction, path-quality evaluation metrics, Gaussian expert-label generation, and the dataset-generation pipeline.

The generated training, validation, and test datasets are not included in the repository because they can be reproduced directly by running `notebooks/03_dataset_generation.ipynb`. Excluding these generated files keeps the repository lightweight while preserving full reproducibility of the prototype experiments.

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